

Minkyu Je

associate professor

基本信息

姓名: Minkyu Je

所属院校: 韩国科学技术院

地区: South Korea

年龄范围: 50岁以上

海外经历: 有

企业经历: 有

学科领域

生物医学工程

集成电路设计与集成系统

研究领域

生物医学传感与检测技术

脑机接口与神经工程

模拟与射频IC设计

低功耗电路与能量管理

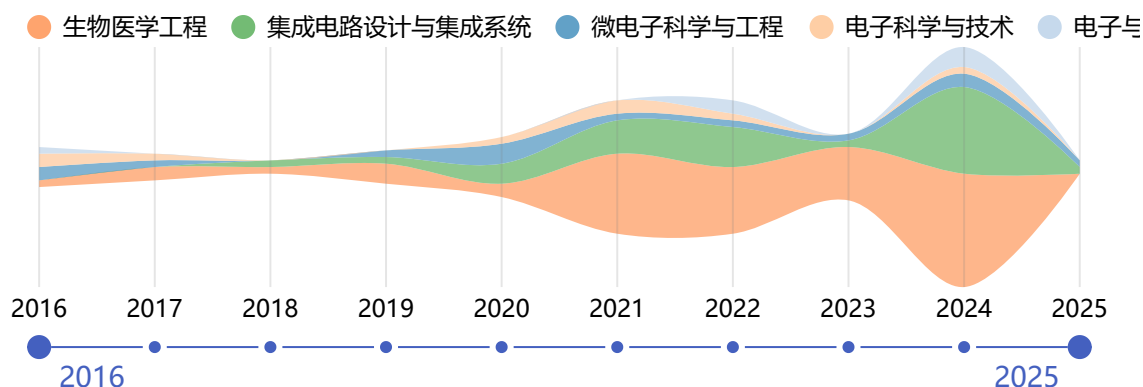
混合信号集成技术

生物信号处理与模式识别

个人简介

Minkyu Je is an Associate Professor at the School of Electrical Engineering, KAIST. He holds a B.S., M.S., and Ph.D. from KAIST. His professional experience spans institutions in Korea and Singapore. His research interests include smart sensor interface ICs, ultra - low - power wireless communication ICs, etc. He has made major research achievements in analog & mixed - signal circuit designs, next - generation wireless IC solutions, etc. His work contributes to UN SDGs. He has a significant research output, including 208 conference contributions, 133 articles, etc., with recent publications in top journals and conferences.

近年发表的论文



Voltage-Summation-Based Compute-in-Memory Technology with Capacitive Synaptic Devices

ADVANCED INTELLIGENT SYSTEMS

Kim, Jung Nam; Kim, Yong Woo; Kim, Boram; Kim, Doo-Hyun; Lee, Gil Sung; Chae, Dong-Hyuk; Je, Minkyu; Koo, Minsuk; Kim, Yoon

被引用次数: 0

2025

A 15.4-ENOB, Fourth-Order Truncation-Error- Shaping NS-SAR-Nested $\Sigma\Delta$ Modulator With Boosted Input Impedance and Range for Biosignal Acquisition

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Kyeongwon; Ha, Sohmyung; Je, Minkyu

被引用次数: 5

2024

A Low-power $\Delta\text{-}\Sigma$ -based Bio-impedance Readout IC with Capacitive-feedback Baseline Cancellation

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Choi, Haidam; Suh, Ji-Hoon; Yun, Gichan; Oh, Sein; Cheon, Song-I; Ha, Sohmyung; Je, Minkyu

被引用次数: 0

2024

Robust High-G Microaccelerometers with Mechanical Low-Pass Filters and Electrical ROIC

2024 IEEE INTERNATIONAL SYMPOSIUM ON INERTIAL SENSORS AND SYSTEMS, INERTIAL

Choi, Yebin; Seok, Minho; Cheon, Song-I; Je, Minkyu; Cho, Young-Ho; Song, Sehwan; Yoon, Sang-Hee; Kim, Kilyug; Uhm, Won-Young; Jang, Junyong; Cho, Yongjun; Jung, Myung-Suk

被引用次数: 0

2024

A High-throughput Impedance Measurement IC Using Synchronous Cyclic Integration Technique

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Ellahi, Karam; Kweon, Soon-Jae; Malik, Asra; Akram, Muhammad Abrar; Cheon, Song-I; Jung, Yoontae; Je, Minkyu; Cheema, Hammad M.; Ha, Sohmyung

被引用次数: 0

2024

A 72-channel Resistive-sensor Interface IC with High Energy Efficiency and a Wide Input Range

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Han, Sunglim; Seong, Hoyong; Oh, Sein; Koo, Jimin; Jin, Hanbit; Kim, Hye Jin; Hai, Sohmyung; Je, Minkyu

被引用次数: 0

2024

A Biopotential Recording IC with <10-ms-Settling Hybrid DC Servo Loop

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Kim, Yegeun; Seok, Changhun; Jung, Yoontae; Ha, Sohmyung; Je, Minkyu

被引用次数: 0

2024

Skew-CIM: Process-Variation-Resilient and Energy-Efficient Computation-in-Memory Design Technique With Skewed Weights

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yi, Donghyeon; Lee, Seoyoung; Choi, Injun; Yun, Gichan; Choi, Edward Jongyoon; Park, Jonghee; Kwak, Jonghoon; Jang, Sung-Joon; Ha, Sohmyung; Chang, Ik-Joon; Je, Minkyu

被引用次数: 1

2024

A Fully Integrated Dynamic-Voltage-Scaling Stimulator IC with Miniaturized Reconfigurable Supply Modulator and Channel Drivers for Cochlear Implants

2024 IEEE CUSTOM INTEGRATED CIRCUITS CONFERENCE, CICC

Kim-Hoang Nguyen; Quyet Nguyen; Quynh-Trang Nguyen; Thanh-Tung Vu; Ahn, Woojin; Loan Pham-Nguyen; Hanh-Phuc Le; Je, Minkyu

被引用次数: 0

2024

A Reconfigurable Multimodal Sensor Interface IC Based on Direct-Conversion $\Delta\Sigma$ Modulator Structure

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Koo, Jimin; Jung, Yoontae; Oh, Sein; Han, Sunglim; Ha, Sohmyung; Je, Minkyu

被引用次数: 0

2024

An Area-Efficient, DC-Coupled VCO-Based CT $\Delta\Sigma$ M with Input-TR-DAC for Neural Recording

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Lee, Woobean; Jung, Yoontae; Jeon, Hyuntak; Koo, Jimin; Oh, Sein; Kweon, Soon-Jae; Je, Minkyu

被引用次数: 0

2024

Ultrahigh-G Accelerometer Readout IC with Adaptive Gain Path for Shock Resilience

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Cheon, Song-I; Park, Seonghyun; Choi, Haidam; Choi, Yebin; Seok, Minho; Cho, Young-Ho; Ha, Sohmyung; Je, Minkyu

被引用次数: 0

2024

A Hybrid High -voltage Regulating Charge Pump for Hectrokinetic Concentration

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Aberra, Aida; Akram, Muhammad Abrar; Kweo, Soon-Jae; Kim, Jongmin; Nguyen, Kim-Hoang; Yun, Gichan; Je, Minkyu; Song, Yong-Ak; Ha, Sohmyung

被引用次数: 0

2024

An Ultrasound Receiver With Bandwidth-Enhanced Current Conveyor and Element-Level Ultrasound Transmitter for Ultrasound Imaging Systems

IEEE SOLID-STATE CIRCUITS LETTERS

Yun, Gichan; Jeong, Kyeongwon; Choi, Haidam; Nam, Seunghyun; Oh, Chaerin; Lee, Hyunjoo Jenny; Ha, Sohmyung; Je, Minkyu

被引用次数: 1

2024

A 187-dB FoMS Power-Efficient Second-Order Highpass $\Delta\Sigma$ Capacitance-to-Digital Converter

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jung, Yoontae; Oh, Sein; Ha, Sohmyung; Je, Minkyu

被引用次数: 0

2024

A Bio-Impedance Readout IC With Complex-Domain Noise-Correlated Baseline Cancellation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, Haidam; Cheon, Song-, I; Yun, Gichan; Oh, Sein; Suh, Ji-Hoon; Ha, Sohmyung; Je, Minkyu

被引用次数: 0

2024

A High-throughput Impedance Measurement IC with Baseline-Canceling Peak Detector

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Malik, Asra; Kweon, Soon-Jae; Ellahi, Karam; Akram, Muhammad Abrar; Cheon, Song-I; Jung, Yoontae; Je, Minkyu; Cheema, Lammad M.; Ha, Sohmyung

被引用次数: 0

2024

Algorithm-Hardware Co-Design for Wearable BCIs: An Evolution from Linear Algebra to Transformers

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Park, Sunyoung; Byun, Wooseok; Je, Minkyu; Kim, Ji-Hoon

被引用次数: 0

2024

A Fully Integrated Dynamic-Voltage-Scaling Stimulator IC for Cochlear Implants

IEEE SOLID-STATE CIRCUITS LETTERS

Nguyen, Kim-Hoang; Nguyen, Quyet; Nguyen, Quynh-Trang; Vu, Thanh-Tung; Ahn, Woojin; Pham-Nguyen, Loan; Le, Hanh-Phuc; Je, Minkyu

被引用次数: 0

2024

A 5.3- μ W 80.8-dB SNDR LNA-Embedded EF-CIFF Third-Order Noise-Shaping SAR ADC for Closed-Loop Neural Recording

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Kim, Yegeun; Seok, Changhun; Jeong, Kyeongwon; Choi, Haidam; Choe, Heewon; Ha, Sohmyung; Je, Minkyu

被引用次数: 2

2024

A Wide-Dynamic-Range, DC-Coupled, Time-Based Neural-Recording IC With Optimized CCO Frequency

IEEE ACCESS

Youn, Donghyun; Kim, Youngin; Choi, Injun; Jung, Yoontae; Jeon, Hyuntak; Lee, Kyungtae; Kweon, Soon-Jae; Ha, Sohmyung; Je, Minkyu

AR-CIM: A 460.22TOPS/W SRAM-based Analog Reconfigurable Computing-in-Memory Macro with 1/2/4/8-Bit Variable Precision

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Choi, Byeongseon; Choi, Edward; Yil, Donghyeon; Chun, Jiho; Ha, Sohmyung; Chang, Ik-Joon; Je, Minkyu

被引用次数: 0

2024

Battery-Free, Wireless Multi-Modal Sensor, and Actuator Array System for Pressure Injury Prevention

SMALL

Han, Hyeonseok; Park, Hyunwoo; Cho, Seokjoo; Lee, Sung-Uk; Choi, Jungrak; Ha, Ji-Hwan; Park, Jaeho; Jung, Young; Kim, Hyunjin; Ahn, Junseong; Kwon, Yeong Jae; Oh, Yong Suk; Je, Minkyu; Park, Inkyu

被引用次数: 1

2024

Impedance-Readout Integrated Circuits for Electrical Impedance Spectroscopy: Methodological Review

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Cheon, Song-, I; Choi, Haidam; Kang, Hyoju; Suh, Ji-Hoon; Park, Seonghyun; Kweon, Soon-Jae; Je, Minkyu; Ha, Sohmyung

被引用次数: 4

2024

A Wireless and Wearable Body-Pressure-Monitoring System for the Prevention of Pressure-Induced Skin Injuries

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Park, Hyunwoo; Kim, Kyuyoung; Kweon, Soon-Jae; Gul, Osman; Choi, Jungrak; Oh, Yong Suk; Park, Inkyu; Je, Minkyu

被引用次数: 1

2023

Optimal Parameter Design of DAC-Based Sinusoidal Signal Generators for Electrical Impedance Spectroscopy

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Cheon, Song-, I; Kweon, Soon-Jae; Aberra, Aida; Je, Minkyu; Ha, Sohmyung

被引用次数: 1

2023

A High-Efficiency Single-Mode Dual-Path Buck-Boost Converter With Reduced Inductor Current

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Cho, Donghee; Cho, Hyungjoo; Oh, Sein; Jung, Yoontae; Ha, Sohmyung; Kim, Chul; Je, Minkyu

被引用次数: 8

2023

A Wide-Bandwidth Ultrasound Receiver and On-Chip Ultrasound Transmitter for Ultrasound Capsule Endoscopy

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeong, Kyeongwon; Yun, Gichan; Choi, Jaesuk; Choi, Injun; Son, Jeehoon; Hwang, Jae Youn; Ha, Sohmyung; Je, Minkyu

被引用次数: 4

2023

A Process-Scalable Ultra-Low-Voltage Sleep Timer With a Time-Domain Amplifier and a Switch-Less Resistance Multiplier

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jung, Chongsoo; Seong, Hoyong; Choi, Injun; Ha, Sohmyung; Je, Minkyu

被引用次数: 0

2023

An Auto-Reconfigurable Multi-Output Regulating Switched-Capacitor DC-DC Converter for Wireless Power Reception and Distribution in Multi-Unit Implantable Devices

IEEE OPEN JOURNAL OF THE SOLID-STATE CIRCUITS SOCIETY

Lee, Unbong; Jung, Wanyoung; Ha, Sohmyung; Je, Minkyu

被引用次数: 2

2023

A Neural Stimulator IC with Dynamic Voltage Scaling Supply and Energy Recycling for Cochlear Implant in Standard 180nm CMOS Process

2023 20TH INTERNATIONAL SOC DESIGN CONFERENCE, ISOC

Kim-Hoang Nguyen; Ahn, Woojin; Je, Minkyu; Quyet Nguyen; Quynh-Trang Nguyen; Thanh-Tung Vu; Loan Pham-Nguyen

被引用次数: 0

2023

Battery-Free, Wireless, Cuff-Type, Multimodal Physical Sensor for Continuous Temperature and Strain Monitoring of Nerve

SMALL

Kim, Seunghwan; Oh, Yong Suk; Lee, Kwanghyoung; Kim, Seongchan; Maeng, Woo-Youl; Kim, Kyung Su; Kim, Ga-Been; Cho, Seokjoo; Han, Hyeonseok; Park, Hyunwoo; Wang, Mengqiu; Avila, Raudel; Xie, Zhaoqian; Ko, Kabseok; Choi, Jungrak; Je, Minkyu; Lee, Hyojin; Lee, Sungho; Koo, Jahyun; Park, Inkyu

被引用次数: 3

2023

Wireless, multimodal sensors for continuous measurement of pressure, temperature, and hydration of patients in wheelchair

NPJ FLEXIBLE ELECTRONICS

Cho, Seokjoo; Han, Hyeonseok; Park, Hyunwoo; Lee, Sung-Uk; Kim, Jae-Hwan; Jeon, Sung Woo; Wang, Mengqiu; Avila, Raudel; Xi, Zhaoqian; Ko, Kabseok; Park, Minsu; Lee, Jungyup; Choi, Myungwoo; Lee, Je-Sang; Min, Weon Gi; Lee, Byeong-Ju; Lee, Soyeong; Choi, Jungrak; Gu, Jimin; Park, Jaeho; Kim, Min Seong; Ahn, Junseong; Gul, Osman; Han, Chankyu; Lee, Gihun; Kim, Seunghwan; Kim, Kyuyoung; Kim, Jeonghyun; Kang, Chang-Mo; Koo, Jahyun; Kwak, Sung Soo; Kim, Sungbong; Choi, Dong Yun; Jeon, Seokwoo; Sung, Hyung Jin; Park, Yong Bae; Je, Minkyu; Cho, Young Tae; Oh, Yong Suk; Park, Inkyu

被引用次数: 23

2023

Battery-Free, Wireless, Ionic Liquid Sensor Arrays to Monitor Pressure and Temperature of Patients in Bed and Wheelchair

SMALL

Han, Hyeonseok; Oh, Yong Suk; Cho, Seokjoo; Park, Hyunwoo; Lee, Sung-Uk; Ko, Kabseok; Park, Jae-Man; Choi, Jungrak; Ha, Ji-Hwan; Han, Chankyu; Zhao, Zichen; Liu, Zhuangjian; Xie, Zhaoqian; Lee, Je-Sang; Min, Weon Gi; Lee, Byeong-Ju; Koo, Jahyun; Choi, Dong Yun; Je, Minkyu; Sun, Jeong-Yun; Park, Inkyu

被引用次数: 12

2023

Inverter-Based Pseudo-Differential Reconfigurable Pixel Circuit Array for Multimodal and High-Density Neural Recordings

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Lee, Taeju; Je, Minkyu

被引用次数: 0

2022

An Impedance Readout IC with Ratio-Based Measurement Techniques for Electrical Impedance Spectroscopy

SENSORS

Cheon, Song-, I; Kweon, Soon-Jae; Kim, Youngin; Koo, Jimin; Ha, Sohmyung; Je, Minkyu

被引用次数: 10

2022

A Multimodal Neural-Recording IC With Reconfigurable Analog Front-Ends for Improved Availability and Usability for Recording Channels

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Lee, Taeju; Kim, Mi Kyung; Lee, Hyunjoo Jenny; Je, Minkyu

被引用次数: 7

2022

On-Chip Sinusoidal Signal Generators for Electrical Impedance Spectroscopy: Methodological Review

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Kweon, Soon-Jae; Rafi, Ayesha Kajol; Cheon, Song-, I; Je, Minkyu; Ha, Sohmyung

被引用次数: 16

2022

A PVT-Robust AFE-Embedded Error-Feedback Noise-Shaping SAR ADC With Chopper-Based Passive High-Pass IIR Filtering for Direct Neural Recording

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Jeong, Kyeongwon; Jung, Yoontae; Yun, Gichan; Youn, Donghyun; Jo, Yehhyun; Lee, Hyunjoo Jenny; Ha, Sohmyung; Je, Minkyu

被引用次数: 6

2022

An Intra-Body Power Transfer System With >1-mW Power Delivered to the Load and 3.3-V DC Output at 160-cm of on-Body Distance

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Cho, Hyungjoo; Suh, Ji-Hoon; Kim, Chul; Ha, Sohmyung; Je, Minkyu

被引用次数: 11

2022

A Wide-Dynamic-Range Neural-Recording IC With Automatic-Gain-Controlled AFE and CT Dynamic-Zoom $\Delta\Sigma$ ADC for Saturation-Free Closed-Loop Neural Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jung, Yoontae; Kweon, Soon-Jae; Jeon, Hyuntak; Choi, Injun; Koo, Jimin; Kim, Mi Kyung; Lee, Hyunjoo Jenny; Ha, Sohmyung; Je, Minkyu

被引用次数: 11

2022

An Energy-Efficient Domain-Specific Reconfigurable Array Processor With Heterogeneous PEs for Wearable Brain-Computer Interface SoCs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Byun, Wooseok; Je, Minkyu; Kim, Ji-Hoon

被引用次数: 5

2022

A 46-nF/10-M ω Range 114-aF/0.37- ω Resolution Parasitic- and Temperature-Insensitive Reconfigurable RC-to-Digital Converter in 0.18- μ m CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

George, Arup K.; Shim, Wooyoon; Kung, Jaeha; Kim, Ji-Hoon; Je, Minkyu; Lee, Junghyup

被引用次数: 4

2022

Advances in Wearable Brain-Computer Interfaces From an Algorithm-Hardware Co-Design Perspective

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Byun, Wooseok; Je, Minkyu; Kim, Ji-Hoon

被引用次数: 6

2022

Dynamic-Range-Enhancement Techniques for Artifact-Tolerant Biopotential-Acquisition ICs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jung, Yoontae; Kweon, Soon-Jae; Lee, Taeju; Jeong, Kyeongwon; Ha, Sohmyung; Je, Minkyu

被引用次数: 3

2022

An SRAM-Based Hybrid Computation-in-Memory Macro Using Current-Reused Differential CCO

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Choi, Injun; Choi, Edward Jongyoon; Yi, Donghyeon; Jung, Yoontae; Seong, Hoyong; Jeon, Hyuntak; Kweon, Soon-Jae; Chang, Ik-Joon; Ha, Sohmyung; Je, Minkyu

被引用次数: 6

2022

Energy-Efficient High-Voltage Pulsers for Ultrasound Transducers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Choi, Jaesuk; Youn, Sangyeon; Hwang, Jae Youn; Ha, Sohmyung; Kim, Chul; Je, Minkyu

被引用次数: 11

2021

A 99.5dB-DR 5kHz-BW Closed-Loop Neural-Recording IC based on Continuous- Time Dynamic-Zoom $\Delta\Sigma$ ADC with Automatic AFE-Gain Control

IEEE ASIAN SOLID-STATE CIRCUITS CONFERENCE (A-SSCC 2021)

Jung, Yoontae; Kweon, Soon-Jae; Jeon, Hyuntak; Lee, Taeju; Choi, Injun; Jeong, Kyeongwon; Kim, Mi Kyung; Lee, Hyunjoo Jenny; Ha, Sohmyung; Je, Minkyu

被引用次数: 2

2021

A Neural Stimulation IC Based on a Reconfigurable Current DAC with In-Situ Neural Recording Function for Cochlear Implant Systems

IEEE ASIAN SOLID-STATE CIRCUITS CONFERENCE (A-SSCC 2021)

Ahn, Woojin; Kim, Doohee; Park, Jonghyeok; Park, Jeong Hoan; Lee, Taeju; Jeong, Kyeongwon; Min, Kyou Sik; Lee, Hoseung; Je, Minkyu

被引用次数: 1	2021
A Load-Current-Regulating OLED Lamp Driver Using a Hybrid Step-Up Converter with 93.21% Efficiency at a High Conversion Ratio of 4.1	
ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)	
Cho, Donghee; Shin, Hongseok; Park, Hyunwoo; Oh, Sein; Lee, Taeju; Ha, Sohmyung; Kim, Chul; Je, Minkyu	
被引用次数: 2	2021
A Scalable Readout IC Based on Wideband Noise Cancelling for Full-Rate Scanning of High-Density Microelectrode Arrays	
2021 43RD ANNUAL INTERNATIONAL CONFERENCE OF THE IEEE ENGINEERING IN MEDICINE & BIOLOGY SOCIETY (EMBC)	
Kim, Jinuk; Shin, Hongseok; Kwon, Soon-Jae; Lee, Seongwook; Ha, Sohmyung; Je, Minkyu	
被引用次数: 1	2021
A High-Precision Single-Ended-Current-to-Differential-Voltage Converter for Reconfigurable Neural Recording Front-Ends	
2021 IEEE INTERNATIONAL MIDWEST SYMPOSIUM ON CIRCUITS AND SYSTEMS (MWSCAS)	
Lee, Taeju; Je, Minkyu	
被引用次数: 2	2021
A Multimodal Neural Activity Readout Integrated Circuit for Recording Fluorescence and Electrical Signals	
IEEE ACCESS	
Lee, Taeju; Park, Jee-Ho; Chou, Namsun; Cho, Il-Joo; Kim, Seong-Jin; Je, Minkyu	
被引用次数: 6	2021
A Precise Lesion Localization System Using a Magnetometer With Real-Time Baseline Cancellation for Laparoscopic Surgery	
IEEE ACCESS	
Kwon, Soon-Jae; Yun, Woojin; Park, Hyunwoo; Park, Jeong-Ho; Lee, Jung Hoon; Lee, Jin; Je, Minkyu; Ha, Sohmyung; Kim, Choul-Young	
被引用次数: 0	2021
Multimodal Neural Interface Circuits for Diverse Interaction With Neuronal Cell Population in Human Brain	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Lee, Taeju; Je, Minkyu	
被引用次数: 6	2021
A Neural Recording and Stimulation Chip with Artifact Suppression for Biomedical Devices	
JOURNAL OF HEALTHCARE ENGINEERING	
Liu, Xu; Li, Juzhe; Chen, Tao; Wang, Wensi; Je, Minkyu	
被引用次数: 2	2021
Logic Device Based on Skyrmion Annihilation	

IEEE TRANSACTIONS ON ELECTRON DEVICES

Song, Moojune; Park, Min Gyu; Ko, San; Jang, Sung Kyu; Je, Minkyu; Kim, Kab-Jin

被引用次数: 27

2021

A Polar-Demodulation-Based Impedance-Measurement IC Using Frequency-Shift Technique With Low Power Consumption and Wide Frequency Range

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Cheon, Song-, I; Kweon, Soon-Jae; Kim, Youngin; Koo, Jimin; Ha, Sohmyung; Je, Minkyu

被引用次数: 9

2021

A Wireless Power and Data Transfer IC for Neural Prostheses Using a Single Inductive Link With Frequency-Splitting Characteristic

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Park, Yechan; Koh, Seok-Tae; Lee, Jeongeun; Kim, Hongkyun; Choi, Jaesuk; Ha, Sohmyung; Kim, Chul; Je, Minkyu

被引用次数: 22

2021

Subthreshold electrical stimulation as a low power electrical treatment for stroke rehabilitation

SCIENTIFIC REPORTS

Kim, Kyungsoo; Yoo, Seung-Jun; Kim, So Yeon; Lee, Taeju; Lim, Sung-Ho; Jang, Jae-Eun; Je, Minkyu; Moon, Cheil; Choi, Ji-Woong

被引用次数: 4

2021

An Ultra-Low-Noise Swing-Boosted Differential Relaxation Oscillator in 0.18- μ m CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Junghyup; George, Arup K.; Je, Minkyu

被引用次数: 29

2020

A 1.0 V, 5.4 pJ/bit GFSK Demodulator Based on an Injection Locked Ring Oscillator for Low-IF Receivers

IEEE ACCESS

Hong, Soonyoung; George, Arup K.; Im, Donggu; Je, Minkyu; Lee, Junghyup

被引用次数: 3

2020

A Multi-Mode ULP Receiver Based on an Injection-Locked Oscillator for IoT Applications

IEEE ACCESS

Hong, Soonyoung; Lee, Sehwan; Lee, Junghyup; Je, Minkyu

被引用次数: 6

2020

Radiation-Hardened Sensor Interface Circuit for Monitoring Severe Accidents in Nuclear Power Plants

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Jeon, Hyuntak; Kwon, Inyong; Je, Minkyu

被引用次数: 11

2020

A Power-Efficient Radiation Sensor Interface with a Peak-Triggered Sampling Scheme for Mobile Dosimeters

SENSORS

Jeon, Hyuntak; Choi, Injun; Kweon, Soon-Jae; Je, Minkyu

被引用次数: 4

2020

Design of Reconfigurable Time-to-Digital Converter Based on Cascaded Time Interpolators for Electrical Impedance Spectroscopy

SENSORS

Shin, Sounghun; Jung, Yoontae; Kweon, Soon-Jae; Lee, Eunseok; Park, Jeong-Ho; Kim, Jinuk; Yoo, Hyung-Joun; Je, Minkyu

被引用次数: 8

2020

A CMRR Enhancement Circuit Employing Gm-Controllable Output Stages for Capacitively Coupled Instrumentation Amplifiers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jeon, Hyuntak; Bang, Jun-Suk; Je, Minkyu

被引用次数: 6

2020

Double-High-Pass-Filter-Based Electrical-Recording Front-Ends and Fluorescence-Recording Front-Ends for Monitoring Multimodal Neural Activity

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Taeju; Je, Minkyu

被引用次数: 7

2020

A High DR, DC-Coupled, Time-Based Neural-Recording IC With Degeneration R-DAC for Bidirectional Neural Interface

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jeon, Hyuntak; Bang, Jun-Suk; Jung, Yoontae; Choi, Injun; Je, Minkyu

被引用次数: 42

2019

Smartphone-based multispectral imaging and machine-learning based analysis for discrimination between seborrheic dermatitis and psoriasis on the scalp

BIOMEDICAL OPTICS EXPRESS

Kim, Sewoong; Kim, Jihun; Hwang, Minjoo; Kim, Manjae; Jo, Seong Jin; Je, Minkyu; Jang, Jae Eun; Lee, Dong Hun; Hwang, Jae Youn

被引用次数: 31

2019

A Sub- μ W/Ch Analog Front-End for Δ -Neural Recording With Spike-Driven Data Compression

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Kim, Seong-Jin; Han, Su-Hyun; Cha, Ji-Hyoung; Liu, Lei; Yao, Lei; Gao, Yuan; Je, Minkyu

被引用次数: 35

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A 1-V 4.6- μ W/channel Fully Differential Neural Recording Front-end IC with Current-controlled Pseudoresistor in 0.18- μ m CMOS

Design of an On-Silicon-Interposer Passive Equalizer for Next Generation High Bandwidth Memory With Data Rate Up To 8 Gb/s

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jeon, Yeseul; Kim, Heegon; Kim, Joungho; Je, Minkyu

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2018

A power-efficient current-mode neural/muscular stimulator design for peripheral nerve prosthesis

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Liu, Xu; Yao, Lei; Ng, Kian Ann; Li, Peng; Wang, Wensi; Je, Minkyu; Xu, Yong Ping

被引用次数: 10

2018

A Resonator-Adaptable Oscillator Using Varactor-Loaded Tuned Amplifiers

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Wang, Yong; Hong, Yan; Goh, Wang Ling; Chai, Kevin T. C.; Je, Minkyu

被引用次数: 5

2017

A novel smart navigation system for intramedullary nailing in orthopedic surgery

PLOS ONE

Choi, Jaesuk; Kim, Jihun; Hwang, Jae Youn; Je, Minkyu; Kim, Jun-Young; Kim, Shin-Yoon

被引用次数: 11

2017

Enhancement of Interface Characteristics of Neural Probe Based on Graphene, ZnO Nanowires, and Conducting Polymer PEDOT

ACS APPLIED MATERIALS & INTERFACES

Ryu, Mingyu; Yang, Jae Hoon; Ahn, Yumi; Sim, Minkyung; Lee, Kyung Hwa; Kim, Kyungsoo; Lee, Taeju; Yoo, Seung-Jun; Kim, So Yeun; Moon, Cheil; Je, Minkyu; Choi, Ji-Woong; Lee, Youngu; Jang, Jae Eun

被引用次数: 53

2017

A 0.8 V Supply- and Temperature-Insensitive Capacitance-to-Digital Converter in 0.18- μ m CMOS

IEEE SENSORS JOURNAL

George, Arup K.; Lee, Junghyup; Kong, Zhi Hui; Je, Minkyu

被引用次数: 20

2016

Smartphone-based multispectral imaging: system development and potential for mobile skin diagnosis

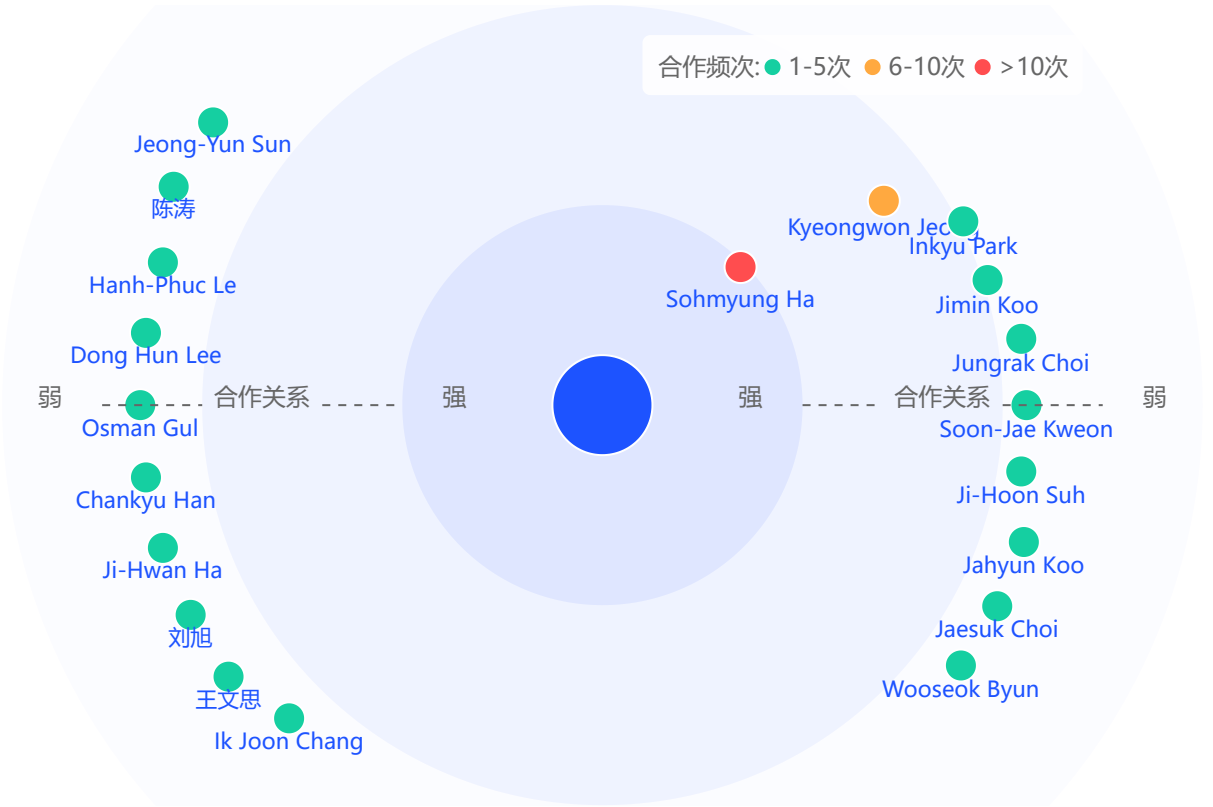
BIOMEDICAL OPTICS EXPRESS

Kim, Sewoong; Cho, Dongrae; Kim, Jihun; Kim, Manjae; Youn, Sangyeon; Jang, Jae Eun; Je, Minkyu; Lee, Dong Hun; Lee, Boreom; Farkas, Daniel L.; Hwang, Jae Youn

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2016

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